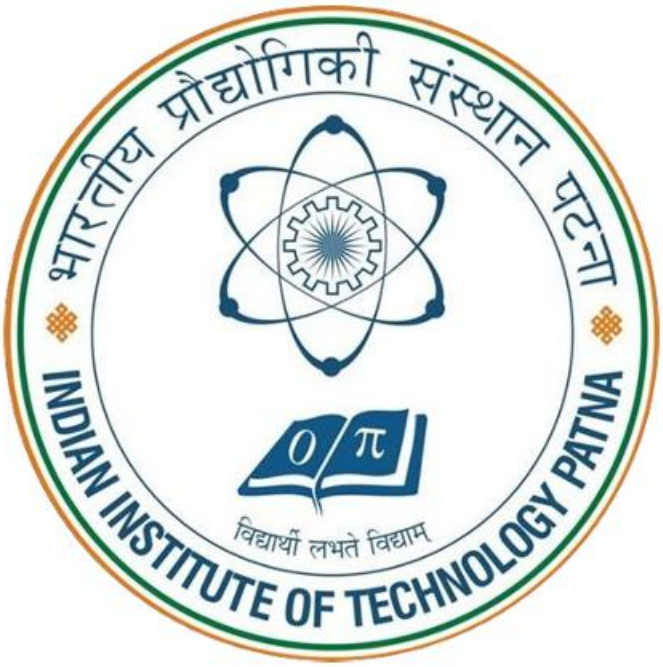
INDIAN INSTITUTE OF TECHNOLOGY, PATNA



CS4113 (IDE-3)

Data Analysis and Visualization

Assignment 1

K-Means Limitations Report

Submitted To:

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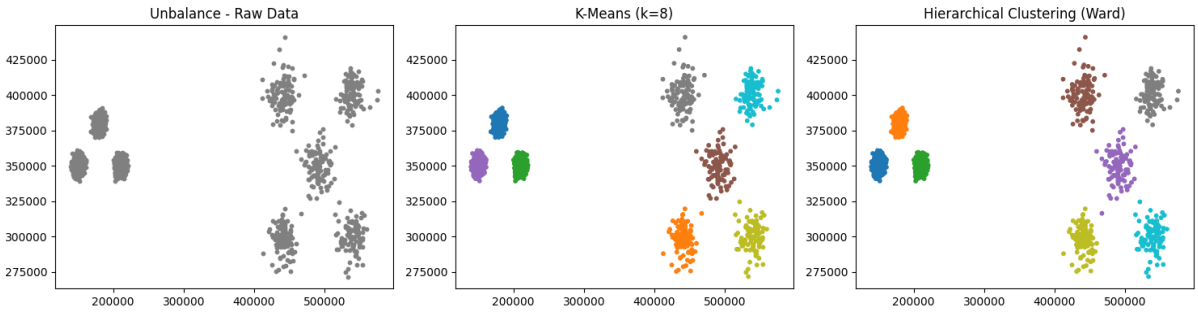
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1. Objective and Method

This study examines three situations in which the assumptions of K-Means clustering become restrictive: clusters with significantly different sizes (Unbalance), clusters with varying densities and non-convex boundaries (Jain), and interwoven, non-globular structures (Spiral). All datasets were standardized before applying the clustering algorithms. K-Means was performed using the benchmark value of **k**. Following tables gives the alternative method used :

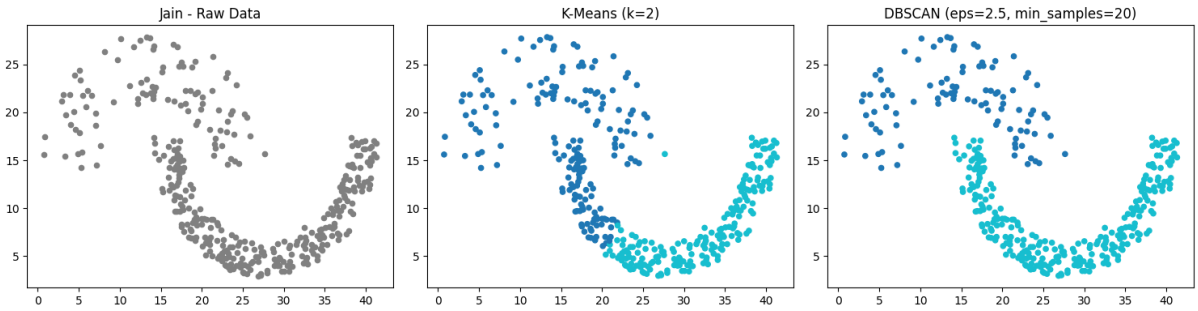
Dataset	**Points**	**K-Means**	**Alternative**	**Settings**
Unbalance	6500	k = 8	Hierarchical	n_clusters=8, linkage='ward'
Jain	373	k = 2	DBSCAN	eps=2.5, min_samples=20
Spiral	312	k = 3	Hierarchical	n_clusters=3, linkage='single'

1.Unbanalce:



Observation: K-Means fails here because it tends to produce clusters of relatively uniform size. When faced with clusters of vastly different sizes, K-Means splits the larger clusters and absorbs smaller ones to minimize within-cluster variance. In contrast, Hierarchical Clustering (with Ward linkage) can successfully isolate the different-sized clusters better, though density-based methods are also excellent for this scenario.

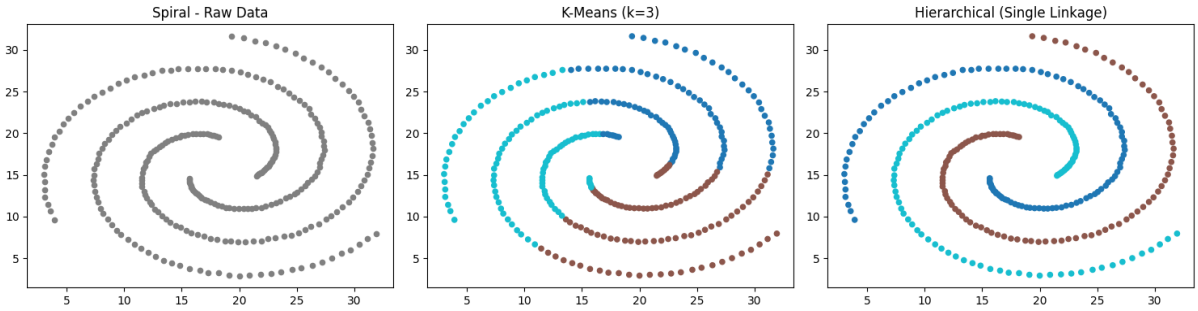
2.Jain:



Observation: K-Means assumes that clusters have similar densities and are roughly spherical, which causes it to split the larger, less dense crescent and merge parts of it

with the denser crescent. DBSCAN successfully clusters this dataset because it groups points based on continuous regions of high density, easily adapting to the varied density and non-globular shape.

3.Spiral:



Observation: K-Means strictly partitions space into Voronoi cells (assuming spherical, convex shapes), meaning it cannot follow the contiguous curves of the interleaved spirals. Hierarchical clustering using single linkage groups points by their nearest neighbours, allowing it to seamlessly follow the non-globular, chaining structure of the spirals and correctly segment them.

Conclusion & General Assumptions of K-Means

Across these three cases, we can conclude that K-Means inherently makes three restrictive assumptions about the underlying data:

1. Clusters are spatially grouped in spherical (convex) shapes.

2. Clusters are of similar sizes (number of points).

3. Clusters share similar densities and variances.

When these assumptions hold, K-Means is extremely fast and efficient. However, when they break down—as demonstrated—K-Means produces highly misleading boundaries.

When to prefer DBSCAN or Hierarchical Clustering over K-Means in practice:

DBSCAN should be preferred when the data contains significant noise/outliers, or when clusters have arbitrary, non-spherical shapes (e.g., spirals, nested circles) but relatively uniform density. It does not require specifying the number of clusters k beforehand.

Hierarchical Clustering (Single Linkage) is excellent for non-globular shapes and discovering hierarchical relationships, but is sensitive to noise.

Hierarchical Clustering (Average Linkage) should be preferred when clusters might have unequal sizes. Ward linkage acts too similarly to K-Means by minimizing variance, but Average Linkage perfectly handles differently sized clusters.

Overall, exploratory data analysis (like a simple scatter plot) should always precede K-Means to ensure the spherical, equal-size assumptions are not wildly violated.